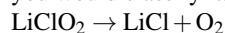


Name: _____

This practice exam covers Chapters 3, 4 of OpenStax Chemistry. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. Classify this reaction as one of the following types: (I) combination, (II) decomposition, (III) single-displacement, (IV) double-displacement (AKA metathesis). You must justify your answer choice. If you believe that the reaction cannot be classified as one of these types, write how you would classify it.



ANSWER: _____

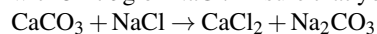
2. Nitric acid is often manufactured in a two-step process from the atmospheric gases nitrogen and oxygen plus hydrogen, prepared by reforming of natural gas. In the first step, nitrogen and hydrogen react to form ammonia. In the second step, ammonia and oxygen react to form nitric acid and water. Write the net chemical equation for the production of nitric acid from nitrogen, hydrogen, and oxygen. Ensure that it is balanced.

ANSWER: _____

3. Calculate the mass of a sample of chlorine monofluoride (ClF) that contains 3.9×10^{19} fluorine atoms.

ANSWER: _____

4. Given this unbalanced chemical equation, calculate and report the theoretical yield of Na_2CO_3 formed by the mixing of 18.31 g of CaCO_3 with 34.70 g of NaCl. Ensure that your answer has the correct number of significant figures.



ANSWER: _____

5. Hydronium bromide reacts with barium sulfide to form barium bromide and hydronium sulfide. Suppose 10.9 g of hydronium bromide is mixed with 7.35 g of barium sulfide. Calculate the maximum mass of barium bromide that could be produced by the chemical reaction. Ensure that your answer has the correct number of significant figures.

ANSWER: _____

6. Calculate the mass percent of sulfur in carbon disulfide (CS_2). Round your answer to the nearest percentage.

ANSWER: _____

7. Of the following labels, identify all that can be applied to the chemical compound H_2SeO_4 : ionic, molecular, acid, and base. Clearly explain why you chose your label(s).

ANSWER: _____

8. Write the net ionic equation for the following chemical reaction, which takes place in aqueous solution:
 $2\text{KOH} + \text{MgCl}_2 \rightarrow \text{Mg}(\text{OH})_2 + 2\text{KCl}$

ANSWER: _____

9. A coin has a mass of $3.430 \times 10^0 \text{ g}$ and an oil tanker has a mass of $7.47 \times 10^8 \text{ kg}$. How many moles of coins have a mass equal to the mass of an oil tanker? Ensure your answer has the correct number of significant figures.

ANSWER: _____

10. Determine whether the following reaction is an oxidation-reduction (redox) reaction. If the reaction is a redox reaction, write down the formula of the reactant that is oxidized and the formula of the reactant that is reduced. Provide an explanation for why you made these choices.
 $\text{CuCr}_2\text{O}_7 + \text{Al} \rightarrow \text{Cu} + \text{Al}_2(\text{Cr}_2\text{O}_7)_3$

ANSWER: _____

11. Sulfuric acid is one of the most widely used chemicals in the world, particularly in the production of fertilizers. It is synthesized in a two-step contact process. In the first step, sulfur (S_8) reacts with oxygen to form sulfur dioxide. In the second step, sulfur dioxide, oxygen gas, and water combine to form sulfuric acid. Write the net chemical equation for the production of sulfuric acid from sulfur, oxygen, and water. Ensure that it is balanced.

ANSWER: _____

12. Which element is 61% (by mass) of diarsenic trisulfide (As_2S_3)? Show your work.

ANSWER: _____

13. Given this unbalanced chemical equation, report the maximum mass of NH_4ClO that could be produced by the mixing of 17.3 g of RbClO with 5.98 g of $(\text{NH}_4)_2\text{S}$. Ensure that your answer has the correct number of significant figures.
 $\text{RbClO} + (\text{NH}_4)_2\text{S} \rightarrow \text{NH}_4\text{ClO} + \text{Rb}_2\text{S}$

ANSWER: _____

14. Eric conducted some measurements and determined that 169. g of bromine liquid participated in a chemical reaction. Calculate the number of moles of bromine liquid that participated in the chemical reaction.

ANSWER: _____

15. Determine whether this reaction will occur. If it does, write the complete balanced equation. If it does not, write "no reaction occurs".
 $\text{Ca}(s) + \text{H}_2\text{O}(l) \rightarrow$

ANSWER: _____

16. A virus has a mass of 5.2×10^{-12} mg and an oil tanker has a mass of 9.504×10^7 kg. How many moles of oil tankers have a mass equal to the mass of a virus? Ensure your answer has the correct number of significant figures.

ANSWER: _____

17. Measurements show that an unknown compound has the following composition:

Element	Mass %
Sulfur	31.1 %
Chlorine	68.9 %

Write the empirical chemical formula of the unknown compound. You must show calculations to support your why you chose your answer.

ANSWER: _____

18. An unknown chemical compound is composed of two unknown elements, A and B. The chemical formula is A_2B_x , where x represents a whole number. Measurements of a sample of the compound show that it contains 1.3 moles of A and 2.0 moles of B. Write the complete chemical formula for the unknown compound.

ANSWER: _____

19. As part of his routine quality-control checks, Robert, an analytical chemist at Southern Union Superb Chemicals Corporation, must determine the concentration of a 2000-L fresh batch of hypochlorous acid solution. He takes a 40.00-mL sample of hypochlorous acid solution and titrates it with a zinc hydroxide solution, which he made by dissolving 8.48 g of solid zinc hydroxide in 1.14 L of distilled water. If Robert finds that it takes 147.5 mL of the zinc hydroxide solution for the titration to reach the equivalence point, what molarity of the hypochlorous acid solution does Robert calculate?

ANSWER: _____

20. Sulfuric acid is one of the most widely used chemicals in the world, particularly in the production of fertilizers. It is synthesized in a two-step contact process. In the first step, sulfur (S_8) reacts with oxygen to form sulfur dioxide. In the second step, sulfur dioxide, oxygen gas, and water combine to form sulfuric acid. Write the net chemical equation for the production of sulfuric acid from sulfur, oxygen, and water. Ensure that it is balanced.

ANSWER: _____

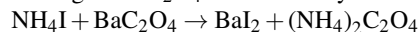
21. Emma must dilute 81.7 mL of a 2.7-mM solution of aqueous iron (III) bromide ($FeBr_3$) until the concentration falls to 0.61 mM. She will do this by adding distilled water to the solution. How much distilled water must she add? Be sure your answer has the correct number of significant figures.

ANSWER: _____

22. Predict the products that form when hydrobromic acid is mixed with sodium hydroxide. Write the balanced chemical equation for this reaction.

ANSWER: _____

23. Given this unbalanced chemical equation, calculate and report the theoretical yield of BaI_2 formed by the mixing of 4.563 g of NH_4I with 6.271 g of BaC_2O_4 . Ensure that your answer has the correct number of significant figures.



ANSWER: _____

24. A strip of solid tin metal is put into a beaker of 0.056-M $AlBr_3$ solution. Does a chemical reaction occur? Explain why it does or why it does not. If it does occur, write the net ionic equation that describes it. Be sure your equation is balanced and has physical-state symbols.

ANSWER: _____

25. Which element is 49% (by mass) of aluminum sulfite ($\text{Al}_2(\text{SO}_3)_3$)? Show your work.

ANSWER: _____

26. The extraction of zinc from its sulfide ores involves a two-step roasting and reduction process. In the first step, zinc sulfide reacts with oxygen to form zinc oxide and sulfur dioxide. In the second step, zinc oxide is reduced by carbon to form zinc metal and carbon monoxide. Write the net chemical equation for the production of zinc from zinc sulfide, oxygen, and carbon. Ensure that it is balanced.

ANSWER: _____

27. Leah prepares a solution of ammonium bromide (NH_4Br) by measuring out 246.4 mmol of ammonium bromide into a 25.-mL volumetric flask and filling the flask to the mark with water. Calculate the concentration of the solution in mol/L. Be sure your answer has the correct number of significant figures.

ANSWER: _____

28. Propane (C_3H_8) can be turned into hydrogen by a two-step reforming process. In the first step, propane and water react to form carbon monoxide and hydrogen. In the second step, carbon monoxide and water react to form hydrogen and carbon dioxide. Write the net chemical equation for the production of hydrogen from propane and water. Ensure that it is balanced.

ANSWER: _____

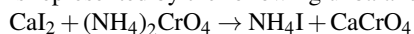
29. Sophia weighs out 0.9019 g of acetic acid ($\text{HC}_2\text{H}_3\text{O}_2$) into a 300.-mL volumetric flask and dilutes the solution to the mark with distilled water. She plans to titrate the acid with 0.341-M sodium hydroxide solution. Calculate the volume of sodium hydroxide solution that Sophia must add to reach the final equivalence point. Ensure your answer has the correct number of significant figures.

ANSWER: _____

30. Phosphoric acid, which most commonly finds uses as a rust inhibitor, a food additive, and an etching agent for dental and orthopedic use, can be synthesized using a two-step thermal process. In the first step, tetratomic phosphorus and diatomic oxygen react to form diphosphorus pentoxide. In the second step, diphosphorus pentoxide and water react to form phosphoric acid. Write the net chemical equation for the production of phosphoric acid from tetratomic phosphorus, diatomic oxygen, and water. Ensure that it is balanced.

ANSWER: _____

31. A way to produce solid calcium chromate is by combining aqueous calcium iodide with aqueous ammonium chromate. The chemical reaction is represented by the following unbalanced chemical equation:



Samantha wants to produce 24.8 mg of calcium chromate. She pours 7.21 mL of a 0.16-M solution of aqueous calcium iodide into a beaker. What volume of a 0.18-M solution of aqueous ammonium chromate is needed to react with the calcium iodide solution to produce 24.8 mg of calcium chromate? If you find that 7.21 mL of the calcium iodide solution is insufficient, report the required volumes of both reactants required to produce the 24.8 mg of calcium chromate.

ANSWER: _____

32. An oil tanker has a mass of 1.789×10^8 kg and a coin has a mass of 3.723×10^0 g. What is the mass of 1 mole of oil tankers? Ensure your answer has the correct number of significant figures.

ANSWER: _____

33. Brandon conducted some measurements and determined that 833. g of iodine solid was produced in a chemical reaction. Calculate the number of moles of iodine solid that was produced in the chemical reaction.

ANSWER: _____

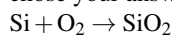
34. Propane (C_3H_8) can be turned into hydrogen by a two-step reforming process. In the first step, propane and water react to form carbon monoxide and hydrogen. In the second step, carbon monoxide and water react to form hydrogen and carbon dioxide. Write the net chemical equation for the production of hydrogen from propane and water. Ensure that it is balanced.

ANSWER: _____

35. Manganese commonly occurs in nature as a mineral. The extraction of manganese from the carbonite mineral rhodochrosite involves a two-step process. In the first step, manganese (II) carbonate and oxygen react to form manganese (IV) oxide and carbon dioxide. In the second step, manganese (IV) oxide and aluminum react to form manganese and aluminum oxide. Write the net chemical equation for the production of manganese from manganese (II) carbonate, oxygen, and aluminum. Ensure that it is balanced.

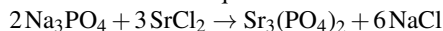
ANSWER: _____

36. As Si is converted into SiO_2 in the following reaction, determine whether silicon is oxidized, reduced, or neither. Clearly explain why you chose your answer. Consider only the silicon found in Si and SiO_2 .



ANSWER: _____

37. Write the net ionic equation for the following chemical reaction, which takes place in aqueous solution:



ANSWER: _____

38. Silver fluoride reacts with sodium sulfide to form sodium fluoride and silver sulfide. Calculate the mass of sodium sulfide needed to produce 5.42 g of sodium fluoride. Ensure your answer has the correct number of significant figures.

ANSWER: _____

39. To help farmers and gardeners, commercial fertilizers have a big three-number "NPK" label on the bag that gives the amounts of three key plant nutrients: nitrogen, phosphorus, and potassium. For example, a bag of "10-10-10" fertilizer has 10% by mass nitrogen, 10% by mass potassium as potash (K_2O) and 10% by mass phosphorus as phosphate (P_2O_5).

ingredient	percent by mass
calcium nitrate ($\text{Ca}(\text{NO}_3)_2$)	27.0%
urea ($(\text{NH}_2)_2\text{CO}$)	35.0%
potassium sulfate (K_2SO_4)	10.0%
ammonium phosphate ($(\text{NH}_4)_3\text{PO}_4$)	21.0%
inert ingredients	7.0%

What should the first (nitrogen) number on the bag's label be? In other words, calculate the percent by mass of nitrogen in this fertilizer. Round your answer to the nearest tenth of a percent.

ANSWER: _____

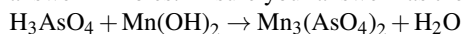
40. Paul is performing a combustion analysis on a compound with a molecular molar mass of 86.18 g/mol that is composed of carbon, hydrogen, and perhaps oxygen. A sample of 6.24 g of the compound is completely burned in excess oxygen and the mass of the products is carefully measured:

Product	Mass (g)
Carbon Dioxide	19.12 g
Water	9.13 g

Use this information to find the molecular formula of the compound.

ANSWER: _____

41. Given this unbalanced chemical equation, report the amount of $\text{Mn}(\text{OH})_2$ needed to produce 0.0932 moles of $\text{Mn}_3(\text{AsO}_4)_2$. Report your answer in moles. Ensure your answer has the correct number of significant figures.

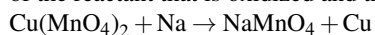


ANSWER: _____

42. A cooking pot has a mass of 9.5×10^{-1} kg and a particular moon has a mass of 4.20×10^{24} kg. What is the mass of 1 mole of this particular moon? Ensure your answer has the correct number of significant figures.

ANSWER: _____

43. Determine whether the following reaction is an oxidation-reduction (redox) reaction. If the reaction is a redox reaction, write down the formula of the reactant that is oxidized and the formula of the reactant that is reduced. Provide an explanation for why you made these choices.



ANSWER: _____

44. As part of her routine quality-control checks, Emily, an analytical chemist at Southern Union Superb Chemicals Corporation, must determine the concentration of a 2000-L fresh batch of hydrofluoric acid solution. She takes a 40.00-mL sample of hydrofluoric acid solution and titrates it with a nickel (II) hydroxide solution, which she made by dissolving 8.26 g of solid nickel (II) hydroxide in 1.358 L of distilled water. If Emily finds that it takes 116.3 mL of the nickel (II) hydroxide solution for the titration to reach the equivalence point, what molarity of the hydrofluoric acid solution does Emily calculate?

ANSWER: _____

45. Christopher has a 2.259-M solution of caesium carbonate (Cs_2CO_3). If he pours 1.65 mL of caesium carbonate into a reaction flask, what is the mass of caesium carbonate that he has added to the flask? Be sure your answer has the correct number of significant figures.

ANSWER: _____

46. Christopher must prepare 780. mL of a 0.29-mM working solution of aqueous strontium iodide (SrI_2). He will do this by pouring some 0.023767-M aqueous strontium iodide stock solution into a graduated cylinder and diluting it with distilled water. How much of the aqueous-strontium iodide stock solution must Christopher pour into the graduated cylinder? Report your answer in mL. Be sure your answer has the correct number of significant figures.

ANSWER: _____

47. Ethan must dilute 20.21 mL of a 5.77-mM solution of aqueous ammonium bromide (NH_4Br) until the concentration falls to 2.46 mM. He will do this by adding distilled water to the solution. How much distilled water must he add? Be sure your answer has the correct number of significant figures.

ANSWER: _____

48. Phosphoric acid, which most commonly finds uses as a rust inhibitor, a food additive, and an etching agent for dental and orthopedic use, can be synthesized using a two-step thermal process. In the first step, tetratomic phosphorus and diatomic oxygen react to form diphosphorus pentoxide. In the second step, diphosphorus pentoxide and water react to form phosphoric acid. Write the net chemical equation for the production of phosphoric acid from tetratomic phosphorus, diatomic oxygen, and water. Ensure that it is balanced.

ANSWER: _____

49. Chromium (III) oxide reacts with ammonium fluoride to form chromium (III) fluoride and ammonium oxide. Calculate the mass of ammonium fluoride needed to produce 9.091 g of ammonium oxide. Ensure your answer has the correct number of significant figures.

ANSWER: _____

50. Nicholas measures the amount of silicon tetrachloride produced during a chemical reaction. He finds that 1259. g is produced. Calculate the number of moles of silicon tetrachloride produced. Ensure your answer has the correct number of significant figures.

ANSWER: _____

51. Each week, Deskens Chemicals, Inc., a manufacturer of fine chemicals, uses 204 kg of compound A and 447 kg of compound B in a reaction that produces only one product, compound C. If the average percent yield of C in the company's reactors is 94.62%, calculate the average mass of compound C that the company actually ships each week.

ANSWER: _____

52. Justin is instructed to dissolve 5.045 g of chromium (III) chloride in 325.4 mL of a 32.6-mM aqueous solution of ammonium phosphate. Calculate the molarity of the chromium (III) cation in the solution after the reaction has occurred. Assume the volume of the solution does not change when chromium (III) is dissolved in it. Ensure your answer has the correct number of significant figures.

ANSWER: _____

53. Emily puts a small sample of Cl_2O_7 in water and lets it dissolve. After equilibrium is reached, she determines that several chemical species remain in the solution. These include Cl_2O_7 and H_2O . Which of the following labels apply to the compound Cl_2O_7 ? Write all that apply. Possible labels include ionic, molecular, strong acid, weak acid, strong base, and weak base. Explain why you chose your answer.

ANSWER: _____

54. Predict the products that form when selenic acid is mixed with calcium hydroxide. Write the balanced chemical equation for this reaction.

ANSWER: _____

55. Rubidium oxide reacts with copper (II) chloride to form rubidium chloride and copper (II) oxide. Suppose 34.10 g of rubidium oxide is mixed with 17.07 g of copper (II) chloride. Calculate and report the theoretical yield of copper (II) oxide formed by this chemical reaction. Ensure that your answer has the correct number of significant figures.

ANSWER: _____

56. Sarah conducted some measurements and determined that 0.7768 g of gold participated in a chemical reaction. Calculate the number of moles of gold that participated in the chemical reaction.

ANSWER: _____

57. Ashley weighs out 1.729 g of boric acid (H_3BO_3) into a 100.-mL volumetric flask and dilutes the solution to the mark with distilled water. She plans to titrate the acid with 0.309-M sodium hydroxide solution. Calculate the volume of sodium hydroxide solution that Ashley must add to reach the final equivalence point. Ensure your answer has the correct number of significant figures.

ANSWER: _____

58. An oil tanker has a mass of 9.5×10^8 kg and an atom of oxygen has a mass of 5.7×10^{-22} g. How many moles of oil tankers have a mass equal to the mass of an atom? Ensure your answer has the correct number of significant figures.

ANSWER: _____

59. The extraction of aluminum metal from the aluminum hydroxide in bauxite ore by the Hall-Héroult process is one of the most remarkable success stories of late-19th-century chemistry, turning aluminum from a rare and precious metal into the cheap commodity it is today. In the first step, aluminum hydroxide reacts to form alumina (which is aluminum oxide) and water. In the second step, alumina and carbon react to form aluminum and carbon dioxide. Write the net chemical equation for the production of aluminum from aluminum hydroxide and carbon. Ensure that it is balanced.

ANSWER: _____

60. Determine whether the following reaction is an oxidation-reduction (redox) reaction. If the reaction is a redox reaction, write down the formula of the reactant that is oxidized and the formula of the reactant that is reduced. Provide an explanation for why you made these choices.
- $$\text{Sr} + \text{Pb}(\text{ClO}_2)_2 \rightarrow \text{Sr}(\text{ClO}_2)_2 + \text{Pb}$$

ANSWER: _____

61. Calculate the mass of a sample of silver carbonate (Ag_2CO_3) that contains 3.022×10^7 carbon atoms.

ANSWER: _____

62. Mary adds 146 mL of a 0.73-M solution of iron (III) sulfate ($\text{Fe}_2(\text{SO}_4)_3$) to a reaction flask. Calculate the amount of iron (III) sulfate that she has added to the flask. Report your answer in moles and be sure your answer has the correct number of significant figures.

ANSWER: _____

63. Nickel (II) hydroxide reacts with hypochlorous acid to form nickel (II) hypochlorite and water. Write the balanced chemical equation for this reaction using the smallest possible whole-number stoichiometric coefficients.

ANSWER: _____

64. Aaron needs to make 109.93 mL of aqueous caesium phosphate (Cs_3PO_4) working solution. He plans to do this by adding distilled water to a 54.35-mL stock solution of 0.020-M aqueous caesium phosphate. What will be the concentration of his working solution? Report your answer in mM. Be sure your answer has the correct number of significant figures.

ANSWER: _____

65. Calculate the mass of a sample of lithium chloride (LiCl) that contains 1.5×10^{18} chlorine atoms.

ANSWER: _____

66. There are two steps in the extraction of copper metal from chalcocite, a copper ore. In the first step, copper (I) sulfide and oxygen gas react to form copper (I) oxide and sulfur dioxide. In the second step, copper (I) oxide and carbon react to form copper and carbon monoxide. Write the net chemical equation for the production of copper from copper (I) sulfide, oxygen, and carbon. Ensure that it is balanced.

ANSWER: _____

67. Magnesium hydroxide reacts with hydrofluoric acid to form water and magnesium fluoride. If 2.840 g of water is produced from the reaction of 17.47 g of magnesium hydroxide and 6.819 g of hydrofluoric acid, calculate the percent yield of water. Ensure that your answer has the correct number of significant figures.

ANSWER: _____

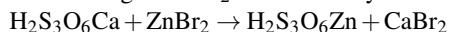
68. Daniel conducted some measurements and determined that 0.742 moles of iodine solid was produced in a chemical reaction. Calculate the mass of iodine solid that was produced in the chemical reaction.

ANSWER: _____

69. Victoria weighs out 1.43 g of an unknown monoprotic acid into a 100.-mL volumetric flask and dilutes the solution to the mark with distilled water. She then titrates this solution with 0.3736-M sodium hydroxide solution. When the titration reaches the equivalence point, Victoria finds that she has added 81.416 mL of the sodium hydroxide solution. Calculate the molar mass of the unknown monoprotic acid. Ensure your answer has the correct number of significant figures.

ANSWER: _____

70. Given this unbalanced chemical equation, report the maximum mass of CaBr_2 that could be produced by the mixing of 16.88 g of $\text{H}_2\text{S}_3\text{O}_6\text{Ca}$ with 28.89 g of ZnBr_2 . Ensure that your answer has the correct number of significant figures.



ANSWER: _____

71. Abigail prepares a solution of potassium chloride (KCl) by measuring out 4.36 g of potassium chloride into a 300.-mL volumetric flask and filling the flask to the mark with water. Calculate the concentration of the solution in mol/L. Be sure your answer has the correct number of significant figures.

ANSWER: _____

72. A coin has a mass of 9.64×10^0 g and a cooking pot has a mass of 3.3×10^{-1} kg. What is the mass of 1 mole of cooking pots? Ensure your answer has the correct number of significant figures.

ANSWER: _____

73. Anna conducted some measurements and determined that 15.64 g of nickel participated in a chemical reaction. Calculate the number of moles of nickel that participated in the chemical reaction.

ANSWER: _____

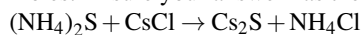
74. Measurements show that an unknown compound has the following composition:

Element	Mass %
B	43.6 %
N	56.4 %

Write the empirical chemical formula of the unknown compound. You must show calculations to support your why you chose your answer.

ANSWER: _____

75. Given this unbalanced chemical equation, report the amount of $(\text{NH}_4)_2\text{S}$ needed to produce 0.0591 moles of NH_4Cl . Report your answer in moles. Ensure your answer has the correct number of significant figures.



ANSWER: _____

76. Sarah has a 0.8712-M solution of ammonium fluoride (NH_4F). If she pours 5.783 mL of ammonium fluoride into a reaction flask, what is the mass of ammonium fluoride that she has added to the flask? Be sure your answer has the correct number of significant figures.

ANSWER: _____

77. In a certain chemical reaction, compound A combines with compound B to produce compound C (and no other products). Measurements were taken of the amounts of A and B present before and after a reaction that produced some C and are presented in the following table. Report the percent yield of C in this reaction to the nearest percent if the actual amount of C that was isolated at the end of the reaction was 36.9g.

Compound	Initial Mass	Final Mass
A	34.9g	0g
B	27.7g	14.7g

ANSWER: _____

78. Propane (C_3H_8) can be turned into hydrogen by a two-step reforming process. In the first step, propane and water react to form carbon monoxide and hydrogen. In the second step, carbon monoxide and water react to form hydrogen and carbon dioxide. Write the net chemical equation for the production of hydrogen from propane and water. Ensure that it is balanced.

ANSWER: _____

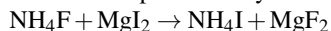
79. Suppose Jonathan heats a sample of a certain barium chlorite hydrate, $\text{Ba}(\text{ClO}_2)_2 \cdot 3\text{H}_2\text{O}$ until all the water is removed and pure $\text{Ba}(\text{ClO}_2)_2$ is left behind. Find the percent by which the mass of the sample decreases in this process.

ANSWER: _____

80. Write the net ionic equation for the following chemical reaction, which takes place in aqueous solution:
 $(\text{NH}_4)_2\text{O} + \text{CuCl}_2 \rightarrow \text{CuO} + 2\text{NH}_4\text{Cl}$

ANSWER: _____

81. A way to produce solid magnesium fluoride is by combining aqueous ammonium fluoride with aqueous magnesium iodide. The chemical reaction is represented by the following unbalanced chemical equation:



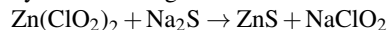
Olivia wants to produce 14.0mg of magnesium fluoride. She pours 6.286mL of a 0.07161-M solution of aqueous ammonium fluoride into a beaker. What volume of a 0.87-M solution of aqueous magnesium iodide is needed to react with the ammonium fluoride solution to produce 14.0mg of magnesium fluoride? If you find that 6.286mL of the ammonium fluoride solution is insufficient, report the required volumes of both reactants required to produce the 14.0mg of magnesium fluoride.

ANSWER: _____

82. As part of his routine quality-control checks, Nicholas, an analytical chemist at Southern Union Superb Chemicals Corporation, must determine the concentration of a 2000-L fresh batch of sulfurous acid solution. He takes a 25-mL sample of sulfurous acid solution and titrates it with a lithium hydroxide solution, which he made by dissolving 2.59 g of solid lithium hydroxide in 1.28 L of distilled water. If Nicholas finds that it takes 662 mL of the lithium hydroxide solution for the titration to reach the equivalence point, what molarity of the sulfurous acid solution does Nicholas calculate?

ANSWER: _____

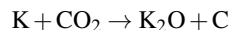
83. A way to produce solid zinc sulfide is by combining aqueous zinc chlorite with aqueous sodium sulfide. The chemical reaction is represented by the following unbalanced chemical equation:



Knowing this, Robert pours 13.97 mL of a 0.022-M solution of aqueous zinc chlorite and then adds 20.5 mL of a 0.153-M solution of aqueous sodium sulfide. He then stirs the mixture and notices zinc sulfide precipitating out of the solution. What is the mass of zinc sulfide that he should expect the reaction to yield?

ANSWER: _____

84. Determine whether the following reaction is an oxidation-reduction (redox) reaction. If the reaction is a redox reaction, write down the formula of the reducing agent and the formula of the oxidizing agent. Provide an explanation for why you made these choices.



ANSWER: _____

85. Ammonium bromide reacts with calcium sulfide to form ammonium sulfide and calcium bromide. Suppose 15.5 g of ammonium bromide is mixed with 4.08 g of calcium sulfide. Calculate and report the theoretical yield of calcium bromide formed by this chemical reaction. Ensure that your answer has the correct number of significant figures.

ANSWER: _____

86. Manganese commonly occurs in nature as a mineral. The extraction of manganese from the carbonite mineral rhodochrosite involves a two-step process. In the first step, manganese (II) carbonate and oxygen react to form manganese (IV) oxide and carbon dioxide. In the second step, manganese (IV) oxide and aluminum react to form manganese and aluminum oxide. Write the net chemical equation for the production of manganese from manganese (II) carbonate, oxygen, and aluminum. Ensure that it is balanced.

ANSWER: _____

87. Measurements show that an unknown compound has the following composition:

Element	Mass %
Hydrogen	7.95 %
Oxygen	42.1 %
Fluorine	50.0 %

Write the empirical chemical formula of the unknown compound.

ANSWER: _____

88. Zoe measures the amount of selenium diiodide produced during a chemical reaction. She finds that 0.6115 mol is produced. Calculate the mass of selenium diiodide produced. Ensure your answer has the correct number of significant figures.

ANSWER: _____

89. Iron is extracted from its oxide ores in a two-step process that has been practiced for centuries. In the first step, iron (III) oxide reacts with carbon monoxide to produce iron (II) oxide and carbon dioxide. In the second step, iron (II) oxide further reacts with carbon monoxide to produce elemental iron and more carbon dioxide. Write the net chemical equation for the production of iron from iron (III) oxide and carbon monoxide. Ensure that it is balanced.

ANSWER: _____

90. Lauren conducted some measurements and determined that 0.602 moles of radium participated in a chemical reaction. Calculate the mass of radium that participated in the chemical reaction.

ANSWER: _____

91. A strip of solid magnesium metal is put into a beaker of 0.29-M $\text{Ca}(\text{BrO}_3)_2$ solution. Does a chemical reaction occur? Explain why it does or why it does not. If it does occur, write the net ionic equation that describes it. Be sure your equation is balanced and has physical-state symbols.

ANSWER: _____

92. Thomas is performing a combustion analysis on a compound with a molecular molar mass of 142.24 g/mol that is composed of carbon, hydrogen, and perhaps oxygen. A sample of 3.66 g of the compound is completely burned in excess oxygen and the mass of the products is carefully measured:

Product	Mass (g)
Carbon Dioxide	10.19 g
Water	4.17 g

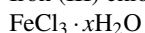
Use this information to find the molecular formula of the compound.

ANSWER: _____

93. Arsenic acid reacts with bismuth hydroxide to form bismuth arsenate and water. Write the balanced chemical equation for this reaction using the smallest possible whole-number stoichiometric coefficients.

ANSWER: _____

94. Iron (III) chloride forms several hydrates with the general form



where x is an integer. If the hydrate is heated, the water can be driven off, leaving pure FeCl_3 behind. Suppose a sample of a certain hydrate of iron (III) chloride is heated until all the water is removed, and a chemist finds that the mass of the sample decreases by 31%. Give the value of x to indicate which hydrate the sample was.

ANSWER: _____

95. Phosphoric acid, which most commonly finds uses as a rust inhibitor, a food additive, and an etching agent for dental and orthopedic use, can be synthesized using a two-step thermal process. In the first step, tetratomic phosphorus and diatomic oxygen react to form diphosphorus pentoxide. In the second step, diphosphorus pentoxide and water react to form phosphoric acid. Write the net chemical equation for the production of phosphoric acid from tetratomic phosphorus, diatomic oxygen, and water. Ensure that it is balanced.

ANSWER: _____

96. Nitric acid is often manufactured in a two-step process from the atmospheric gases nitrogen and oxygen plus hydrogen, prepared by reforming of natural gas. In the first step, nitrogen and hydrogen react to form ammonia. In the second step, ammonia and oxygen react to form nitric acid and water. Write the net chemical equation for the production of nitric acid from nitrogen, hydrogen, and oxygen. Ensure that it is balanced.

ANSWER: _____

97. Emily conducted some measurements and determined that 0.998 moles of dubnium participated in a chemical reaction. Calculate the mass of dubnium that participated in the chemical reaction.

ANSWER: _____

98. Zoe measures the amount of chromium (III) nitrate produced during a chemical reaction. She finds that 1113.1 g is produced. Calculate the number of moles of chromium (III) nitrate produced. Ensure your answer has the correct number of significant figures.

ANSWER: _____

99. Aaron prepares a solution of iron (III) bicarbonate ($\text{Fe}(\text{HCO}_3)_3$) by measuring out 0.261 g of iron (III) bicarbonate into a 50.-mL volumetric flask and filling the flask to the mark with water. Calculate the concentration of the solution in mol/L. Be sure your answer has the correct number of significant figures.

ANSWER: _____

100. Determine whether this reaction will occur. If it does, write the complete balanced equation. If it does not, write "no reaction occurs".
 $\text{Sr}(s) + \text{H}_2\text{O}(l) \rightarrow$

ANSWER: _____

101. Rachel must prepare 608 mL of a 0.64-mM working solution of aqueous potassium sulfide (K_2S). She will do this by pouring some 0.319-M aqueous potassium sulfide stock solution into a graduated cylinder and diluting it with distilled water. How much of the aqueous-potassium sulfide stock solution must Rachel pour into the graduated cylinder? Report your answer in mL. Be sure your answer has the correct number of significant figures.

ANSWER: _____

102. What major chemical species are present in an aqueous solution of 0.013-M iron (III) sulfide?

ANSWER: _____

103. The industrial synthesis of acetic acid (CH_3COOH), the acid found in vinegar, via the methanol carbonylation process involves two key steps. In the first step, methanol (CH_3OH) reacts with carbon monoxide to form methyl formate (HCOOCH_3). In the second step, methyl formate reacts with carbon monoxide and water to produce acetic acid, carbon monoxide, and water. Write the net chemical equation for the production of acetic acid from methanol, carbon monoxide, and water. Ensure that it is balanced.

ANSWER: _____

104. Suppose Nathan heats a sample of a certain calcium phosphate hydrate, $\text{Ca}_3(\text{PO}_4)_2 \cdot 8\text{H}_2\text{O}$ until all the water is removed and pure $\text{Ca}_3(\text{PO}_4)_2$ is left behind. Find the percent by which the mass of the sample decreases in this process.

ANSWER: _____

105. Each week, Deskins Chemicals, Inc., a manufacturer of fine chemicals, uses 525 kg of compound A and 421 kg of compound B in a reaction that produces only one product, compound C. If the average percent yield of C in the company's reactors is 92.42%, calculate the average mass of compound C that the company actually ships each week.

ANSWER: _____

106. Titanium is extracted from its ore, rutile (titanium dioxide), via the two-step Kroll process. In the first step, titanium dioxide is converted to titanium tetrachloride by reacting it with chlorine and carbon. In the second step, titanium tetrachloride is reduced using molten magnesium to form titanium metal and magnesium chloride (which is then often refined back to magnesium). Write the net chemical equation for the production of titanium from titanium dioxide, chlorine, carbon, and magnesium. Ensure that it is balanced.

ANSWER: _____

107. Each week, Deskins Chemicals, Inc., a manufacturer of fine chemicals, uses 350. kg of compound A and 687 kg of compound B in a reaction that produces only one product, compound C. If the average percent yield of C in the company's reactors is 70.16%, calculate the average mass of compound C that the company actually ships each week.

ANSWER: _____

108. Lauren weighs out 0.948 g of an unknown triprotic acid into a 50.-mL volumetric flask and dilutes the solution to the mark with distilled water. She then titrates this solution with 0.388-M strontium hydroxide solution. When the titration reaches the equivalence point, Lauren finds that she has added 37.40 mL of the strontium hydroxide solution. Calculate the molar mass of the unknown triprotic acid. Ensure your answer has the correct number of significant figures.

ANSWER: _____

109. What major chemical species are present in an aqueous solution of 0.71-M magnesium phosphate?

ANSWER: _____

110. Rewrite the following chemical equation as a balanced equation using the smallest possible whole-number stoichiometric coefficients.
 $\text{C}_9\text{H}_{19}\text{OH} + \text{O}_2 \rightarrow \text{H}_2\text{O} + \text{CO}_2$

ANSWER: _____

111. Calculate the number of oxygen atoms in a 1.517-g sample of caesium hydroxide (CsOH).

ANSWER: _____

112. Scott conducted some measurements and determined that 0.644 moles of titanium participated in a chemical reaction. Calculate the mass of titanium that participated in the chemical reaction.

ANSWER: _____

113. In a certain chemical reaction, compound A combines with compound B to produce compound C (and no other products). Measurements were taken of the amounts of A and B present before and after a reaction that produced some C and are presented in the following table. Report the percent yield of C in this reaction to the nearest percent if the actual amount of C that was isolated at the end of the reaction was 41.2 g.

Compound	Initial Mass	Final Mass
A	24.7 g	1.5 g
B	22.6 g	0 g

ANSWER: _____

114. Given this unbalanced chemical equation, report the mass of HClO_4 needed to produce 8.530 g of H_2O . Ensure your answer has the correct number of significant figures.
 $\text{Pb}(\text{OH})_2 + \text{HClO}_4 \rightarrow \text{Pb}(\text{ClO}_4)_2 + \text{H}_2\text{O}$

ANSWER: _____

115. The industrial synthesis of acetic acid (CH_3COOH), the acid found in vinegar, via the methanol carbonylation process involves two key steps. In the first step, methanol (CH_3OH) reacts with carbon monoxide to form methyl formate (HCOOCH_3). In the second step, methyl formate reacts with carbon monoxide and water to produce acetic acid, carbon monoxide, and water. Write the net chemical equation for the production of acetic acid from methanol, carbon monoxide, and water. Ensure that it is balanced.

ANSWER: _____

116. Joshua measures the amount of calcium iodide produced during a chemical reaction. He finds that 674.99 g is produced. Calculate the number of moles of calcium iodide produced. Ensure your answer has the correct number of significant figures.

ANSWER: _____

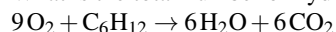
117. Elizabeth adds 119.4 mL of a 0.131-M solution of potassium hydroxide (KOH) to a reaction flask. Calculate the amount of potassium hydroxide that she has added to the flask. Report your answer in moles and be sure your answer has the correct number of significant figures.

ANSWER: _____

118. A particular moon has a mass of 6.4×10^{22} kg and a cooking pot has a mass of 6.69×10^0 kg. How many moles of cooking pots have a mass equal to the mass of this particular moon? Ensure your answer has the correct number of significant figures.

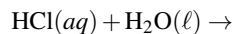
ANSWER: _____

119. What is the total number of hydrogen atoms shown on the product side of this chemical equation?



ANSWER: _____

120. Predict the products of the reaction below by completing the right-hand side of the chemical equation. Provide the full balanced equation with state symbols.



ANSWER: _____

121. Calculate the number of germanium atoms in a 0.0482-g sample of silicon monogermanide (SiGe).

ANSWER: _____

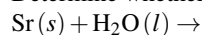
122. Suppose Daniel heats a sample of a certain silver sulfate hydrate, $\text{Ag}_2\text{SO}_4 \cdot 3\text{H}_2\text{O}$ until all the water is removed and pure Ag_2SO_4 is left behind. Find the percent by which the mass of the sample decreases in this process.

ANSWER: _____

123. What major chemical species are present in an aqueous solution of 0.0219-M iron (III) sulfide?

ANSWER: _____

124. Determine whether this reaction will occur. If it does, write the complete balanced equation. If it does not, write "no reaction occurs".



ANSWER: _____

125. Kevin conducted some measurements and determined that 707. g of fluorine gas participated in a chemical reaction. Calculate the number of moles of fluorine gas that participated in the chemical reaction.

ANSWER: _____

126. Aluminum sulfide reacts with hydronium iodide to form hydronium sulfide and aluminum iodide. Write the balanced chemical equation for this reaction using the smallest possible whole-number stoichiometric coefficients.

ANSWER: _____

127. Predict the products that form when nitrous acid is mixed with calcium hydroxide. Write the balanced chemical equation for this reaction.

ANSWER: _____

128. Jason is instructed to dissolve 4.18 g of ammonium sulfite in 76.2 mL of a 62.44-mM aqueous solution of calcium bromide. Calculate the molarity of the calcium cation in the solution after the reaction has occurred. Assume the volume of the solution does not change when calcium is dissolved in it. Ensure your answer has the correct number of significant figures.

ANSWER: _____

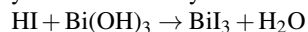
129. The extraction of aluminum metal from the aluminum hydroxide in bauxite ore by the Hall-Héroult process is one of the most remarkable success stories of late-19th-century chemistry, turning aluminum from a rare and precious metal into the cheap commodity it is today. In the first step, aluminum hydroxide reacts to form alumina (which is aluminum oxide) and water. In the second step, alumina and carbon react to form aluminum and carbon dioxide. Write the net chemical equation for the production of aluminum from aluminum hydroxide and carbon. Ensure that it is balanced.

ANSWER: _____

130. Phosphoric acid reacts with aluminum hydroxide to form aluminum phosphate and water. Suppose 25.1 g of phosphoric acid is mixed with 13.8 g of aluminum hydroxide. Calculate the maximum mass of aluminum phosphate that could be produced by the chemical reaction. Ensure that your answer has the correct number of significant figures.

ANSWER: _____

131. Classify this reaction as one of the following types: (I) combination, (II) decomposition, (III) single-displacement, (IV) double-displacement (AKA metathesis). You must justify your answer choice. If you believe that the reaction cannot be classified as one of these types, write how you would classify it.



ANSWER: _____

132. Iron is extracted from its oxide ores in a two-step process that has been practiced for centuries. In the first step, iron (III) oxide reacts with carbon monoxide to produce iron (II) oxide and carbon dioxide. In the second step, iron (II) oxide further reacts with carbon monoxide to produce elemental iron and more carbon dioxide. Write the net chemical equation for the production of iron from iron (III) oxide and carbon monoxide. Ensure that it is balanced.

ANSWER: _____

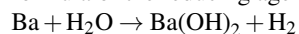
133. Calculate the number of boron atoms in a 89-g sample of boron monoantimonide (BSb).

ANSWER: _____

134. The chemical formula for zinc sulfate is ZnSO_4 . Calculate the molar mass of zinc sulfate. Round your answer to 4 significant figures.

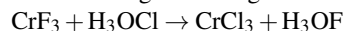
ANSWER: _____

135. Determine whether the following reaction is an oxidation-reduction (redox) reaction. If the reaction is a redox reaction, write down the formula of the reducing agent and the formula of the oxidizing agent. Provide an explanation for why you made these choices.



ANSWER: _____

136. Given this unbalanced chemical equation, report the mass of H_3OCl needed to produce 4.94 g of H_3OF . Ensure your answer has the correct number of significant figures.



ANSWER: _____

137. The industrial synthesis of acetic acid (CH_3COOH), the acid found in vinegar, via the methanol carbonylation process involves two key steps. In the first step, methanol (CH_3OH) reacts with carbon monoxide to form methyl formate (HCOOCH_3). In the second step, methyl formate reacts with carbon monoxide and water to produce acetic acid, carbon monoxide, and water. Write the net chemical equation for the production of acetic acid from methanol, carbon monoxide, and water. Ensure that it is balanced.

ANSWER: _____

138. Measurements show that an unknown compound has the following composition:

Element	Mass %
Zinc	44.95 %
Sulfur	22.04 %
Oxygen	33.00 %

Write the empirical chemical formula of the unknown compound.

ANSWER: _____

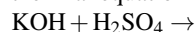
139. The extraction of zinc from its sulfide ores involves a two-step roasting and reduction process. In the first step, zinc sulfide reacts with oxygen to form zinc oxide and sulfur dioxide. In the second step, zinc oxide is reduced by carbon to form zinc metal and carbon monoxide. Write the net chemical equation for the production of zinc from zinc sulfide, oxygen, and carbon. Ensure that it is balanced.

ANSWER: _____

140. Which element is 38% (by mass) of caesium bromide (CsBr)? Show your work.

ANSWER: _____

141. Predict the products of the reaction below. In other words, write the products to complete the right-hand side of the chemical equation. Ensure the final equation is balanced.

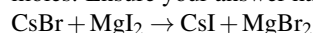


ANSWER: _____

142. Leah measures the amount of germanium ditelluride produced during a chemical reaction. She finds that 0.0934 mol is produced. Calculate the mass of germanium ditelluride produced. Ensure your answer has the correct number of significant figures.

ANSWER: _____

143. Given this unbalanced chemical equation, report the amount of CsBr needed to produce 0.05370 moles of MgBr_2 . Report your answer in moles. Ensure your answer has the correct number of significant figures.



ANSWER: _____

144. Samantha adds 348 mL of a 0.7434-M solution of lithium hypochlorite (LiClO) to a reaction flask. Calculate the amount of lithium hypochlorite that she has added to the flask. Report your answer in moles and be sure your answer has the correct number of significant figures.

ANSWER: _____

145. Of the following labels, identify all that can be applied to the chemical compound HNO_2 : ionic, molecular, acid, and base. Clearly explain why you chose your label(s).

ANSWER: _____

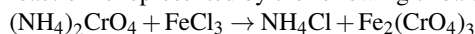
146. Daniel conducted some measurements and determined that 2.72 g of phosphorus participated in a chemical reaction. Calculate the number of moles of phosphorus that participated in the chemical reaction.

ANSWER: _____

147. Manganese commonly occurs in nature as a mineral. The extraction of manganese from the carbonite mineral rhodochrosite involves a two-step process. In the first step, manganese (II) carbonate and oxygen react to form manganese (IV) oxide and carbon dioxide. In the second step, manganese (IV) oxide and aluminum react to form manganese and aluminum oxide. Write the net chemical equation for the production of manganese from manganese (II) carbonate, oxygen, and aluminum. Ensure that it is balanced.

ANSWER: _____

148. A way to produce solid iron (III) chromate is by combining aqueous ammonium chromate with aqueous iron (III) chloride. The chemical reaction is represented by the following unbalanced chemical equation:



Knowing this, Ryan pours 26.5 mL of a 0.2462-M solution of aqueous ammonium chromate and then adds 11.35 mL of a 0.12-M solution of aqueous iron (III) chloride. He then stirs the mixture and notices iron (III) chromate precipitating out of the solution. What is the mass of iron (III) chromate that he should expect the reaction to yield?

ANSWER: _____

149. Hannah measures the amount of aluminum nitrite produced during a chemical reaction. She finds that 0.365 mol is produced. Calculate the mass of aluminum nitrite produced. Ensure your answer has the correct number of significant figures.

ANSWER: _____

150. In a certain chemical reaction, compound A combines with compound B to produce compound C (and no other products). Measurements were taken of the amounts of A and B present before and after a reaction that produced some C and are presented in the following table. Report the percent yield of C in this reaction to the nearest percent if the actual amount of C that was isolated at the end of the reaction was 49.2 g.

Compound	Initial Mass	Final Mass
A	79.1 g	34.8 g
B	27.0 g	0 g

ANSWER: _____

151. Magnesium hydroxide reacts with hydrobromic acid to form magnesium bromide and water. Calculate the moles of hydrobromic acid needed to produce 0.6872 moles of water. Ensure your answer has the correct number of significant figures.

ANSWER: _____

152. Nitric acid is often manufactured in a two-step process from the atmospheric gases nitrogen and oxygen plus hydrogen, prepared by reforming of natural gas. In the first step, nitrogen and hydrogen react to form ammonia. In the second step, ammonia and oxygen react to form nitric acid and water. Write the net chemical equation for the production of nitric acid from nitrogen, hydrogen, and oxygen. Ensure that it is balanced.

ANSWER: _____

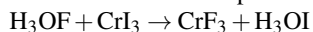
153. Nicholas prepares a solution of lithium hydroxide (LiOH) by measuring out 11.49 mmol of lithium hydroxide into a 100.-mL volumetric flask and filling the flask to the mark with water. Calculate the concentration of the solution in mol/L. Be sure your answer has the correct number of significant figures.

ANSWER: _____

154. Julia must dilute 29.51 mL of a 5.26-mM solution of aqueous hydronium bromide (H_3OBr) until the concentration falls to 3.41 mM. She will do this by adding distilled water to the solution. How much distilled water must she add? Be sure your answer has the correct number of significant figures.

ANSWER: _____

155. A way to produce solid chromium (III) fluoride is by combining aqueous hydronium fluoride with aqueous chromium (III) iodide. The chemical reaction is represented by the following unbalanced chemical equation:



Knowing this, Kyle pours 10.7 mL of a 0.44-M solution of aqueous hydronium fluoride and then adds 15.08 mL of a 0.013-M solution of aqueous chromium (III) iodide. He then stirs the mixture and notices chromium (III) fluoride precipitating out of the solution. What is the mass of chromium (III) fluoride that he should expect the reaction to yield?

ANSWER: _____

156. What is the oxidation number of oxygen in carbon dioxide?

ANSWER: _____

157. Of the following labels, identify all that can be applied to the chemical compound H_2CrO_4 : ionic, molecular, acid, and base. Clearly explain why you chose your label(s).

ANSWER: _____

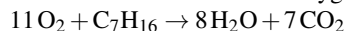
158. Calculate the mass percent of oxygen in dinitrogen trioxide (N_2O_3). Round your answer to the nearest percentage.

ANSWER: _____

159. A brick has a mass of 7.89×10^0 kg and a cooking pot has a mass of 9.538×10^0 kg. How many moles of cooking pots have a mass equal to the mass of a brick? Ensure your answer has the correct number of significant figures.

ANSWER: _____

160. What is the total number of oxygen atoms shown on the product side of this chemical equation?

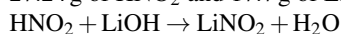


ANSWER: _____

161. Anna conducted some measurements and determined that 0.269 moles of fluorine gas participated in a chemical reaction. Calculate the mass of fluorine gas that participated in the chemical reaction.

ANSWER: _____

162. Given this unbalanced chemical equation, calculate and report the percent yield of H_2O if 8.56 g of H_2O is produced from the reaction of 27.24 g of HNO_2 and 17.7 g of LiOH . Ensure that your answer has the correct number of significant figures.

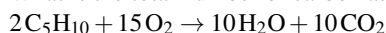


ANSWER: _____

163. A brick has a mass of 2.95×10^0 kg and an oil tanker has a mass of 2.545×10^7 kg. How many moles of bricks have a mass equal to the mass of an oil tanker? Ensure your answer has the correct number of significant figures.

ANSWER: _____

164. What is the total number of carbon atoms shown on the product side of this chemical equation?



ANSWER: _____

165. Titanium is extracted from its ore, rutile (titanium dioxide), via the two-step Kroll process. In the first step, titanium dioxide is converted to titanium tetrachloride by reacting it with chlorine and carbon. In the second step, titanium tetrachloride is reduced using molten magnesium to form titanium metal and magnesium chloride (which is then often refined back to magnesium). Write the net chemical equation for the production of titanium from titanium dioxide, chlorine, carbon, and magnesium. Ensure that it is balanced.

ANSWER: _____

166. Aaron must prepare 652.1 mL of a 0.859-mM working solution of aqueous lithium oxalate ($\text{Li}_2\text{C}_2\text{O}_4$). He will do this by pouring some 0.07314-M aqueous lithium oxalate stock solution into a graduated cylinder and diluting it with distilled water. How much of the aqueous-lithium oxalate stock solution must Aaron pour into the graduated cylinder? Report your answer in mL. Be sure your answer has the correct number of significant figures.

ANSWER: _____

167. Caleb has a 2.12×10^{-3} -M solution of iron (III) chlorite ($\text{Fe}(\text{ClO}_2)_3$). If he needs to add 0.02602 g of iron (III) chlorite to a reaction flask, what volume of the solution should he pour? Be sure your answer has the correct number of significant figures.

ANSWER: _____

168. Measurements show that an unknown compound has the following composition:

Element	Mass %
Potassium	67.3 %
Fluorine	32.7 %

Write the empirical chemical formula of the unknown compound. You must show calculations to support your why you chose your answer.

ANSWER: _____

169. Phosphoric acid reacts with sodium hydroxide to form sodium phosphate and water. Calculate the moles of phosphoric acid needed to produce 1.000 moles of sodium phosphate. Ensure your answer has the correct number of significant figures.

ANSWER: _____

170. A virus has a mass of 3.2×10^{-12} mg and an atom of iron has a mass of 1.4×10^{-21} g. What is the mass of 1 mole of viruses? Ensure your answer has the correct number of significant figures.

ANSWER: _____

171. Christopher is instructed to dissolve 0.9514 g of chromium (III) chloride in 64.2 mL of a 829-mM aqueous solution of lithium sulfide. Calculate the molarity of the sulfide anion in the solution after the reaction has occurred. Assume the volume of the solution does not change when sulfide is dissolved in it. Ensure your answer has the correct number of significant figures.

ANSWER: _____

172. Natalie prepares a solution of rubidium sulfide (Rb_2S) by measuring out 0.0386 g of rubidium sulfide into a 500.-mL volumetric flask and filling the flask to the mark with water. Calculate the concentration of the solution in mol/L. Be sure your answer has the correct number of significant figures.

ANSWER: _____

173. Jessica weighs out 1.009 g of an unknown monoprotic acid into a 500.-mL volumetric flask and dilutes the solution to the mark with distilled water. She then titrates this solution with 0.370-M cobalt (II) hydroxide solution. When the titration reaches the equivalence point, Jessica finds that she has added 21.64 mL of the cobalt (II) hydroxide solution. Calculate the molar mass of the unknown monoprotic acid. Ensure your answer has the correct number of significant figures.

ANSWER: _____

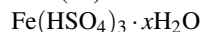
174. SbBr_3 is the formula for antimony tribromide. Emily, a chemist, measures a sample of antimony tribromide and determines that it contains 0.1295 moles of bromine. How many moles of antimony are in the sample?

ANSWER: _____

175. Sulfurous acid reacts with lead (II) hydroxide to form lead (II) sulfite and water. Suppose 4.91 g of sulfurous acid is mixed with 11.1 g of lead (II) hydroxide. Calculate and report the theoretical yield of lead (II) sulfite formed by this chemical reaction. Ensure that your answer has the correct number of significant figures.

ANSWER: _____

176. Iron (III) bisulfate forms several hydrates with the general form



where x is an integer. If the hydrate is heated, the water can be driven off, leaving pure $\text{Fe}(\text{HSO}_4)_3$ behind. Suppose a sample of a certain hydrate of iron (III) bisulfate is heated until all the water is removed, and a chemist finds that the mass of the sample decreases by 13%. Give the value of x to indicate which hydrate the sample was.

ANSWER: _____

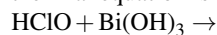
177. HCl is the formula for hydrogen monochloride. Sam, a chemical engineer, measures a sample of hydrogen monochloride and determines that it contains 0.542 moles of hydrogen. How many moles of chlorine are in the sample?

ANSWER: _____

178. The chemical formula for trigermanium tetraantimonide is Ge_3Sb_4 . Calculate the molar mass of trigermanium tetraantimonide. Round your answer to 4 significant figures.

ANSWER: _____

179. Predict the products of the reaction below. In other words, write the products to complete the right-hand side of the chemical equation. Ensure the final equation is balanced.

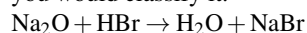


ANSWER: _____

180. The chemical formula for carbon tetrafluoride is CF_4 . Calculate the molar mass of carbon tetrafluoride. Round your answer to 5 significant figures.

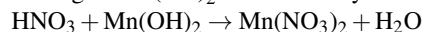
ANSWER: _____

181. Classify this reaction as one of the following types: (I) combination, (II) decomposition, (III) single-displacement, (IV) double-displacement (AKA metathesis). You must justify your answer choice. If you believe that the reaction cannot be classified as one of these types, write how you would classify it.



ANSWER: _____

182. Given this unbalanced chemical equation, calculate and report the theoretical yield of H_2O formed by the mixing of 47.49 g of HNO_3 with 20.17 g of $\text{Mn}(\text{OH})_2$. Ensure that your answer has the correct number of significant figures.



ANSWER: _____

183. To help farmers and gardeners, commercial fertilizers have a big three-number "NPK" label on the bag that gives the amounts of three key plant nutrients: nitrogen, phosphorus, and potassium. For example, a bag of "10-10-10" fertilizer has 10% by mass nitrogen, 10% by mass potassium as potash (K_2O) and 10% by mass phosphorus as phosphate (P_2O_5).

ingredient	percent by mass
calcium nitrate ($\text{Ca}(\text{NO}_3)_2$)	15.0%
urea ($(\text{NH}_2)_2\text{CO}$)	14.0%
potassium sulfate (K_2SO_4)	44.0%
ammonium phosphate ($(\text{NH}_4)_3\text{PO}_4$)	6.0%
inert ingredients	21.0%

What should the third (phosphorus) number on the bag's label be? In other words, calculate the percent by mass of phosphorus in this fertilizer. Round your answer to the nearest tenth of a percent.

ANSWER: _____

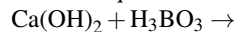
184. A particular planet has a mass of 9.3×10^{28} g and a virus has a mass of 9.09×10^{-11} mg. What is the mass of 1 mole of viruses? Ensure your answer has the correct number of significant figures.

ANSWER: _____

185. William has a 1.89-M solution of magnesium chloride (MgCl_2). If he pours 29.7 mL of magnesium chloride into a reaction flask, what is the mass of magnesium chloride that he has added to the flask? Be sure your answer has the correct number of significant figures.

ANSWER: _____

186. Predict the products of the reaction below. In other words, write the products to complete the right-hand side of the chemical equation. Ensure the final equation is balanced.



ANSWER: _____

187. Victoria needs to make 368.3 mL of aqueous potassium oxide (K_2O) working solution. She plans to do this by adding distilled water to a 66.303-mL stock solution of 0.0976-M aqueous potassium oxide. What will be the concentration of her working solution? Report your answer in mM. Be sure your answer has the correct number of significant figures.

ANSWER: _____

188. A particular moon has a mass of 7.678×10^{23} kg and a cooking pot has a mass of 5.0×10^0 kg. What is the mass of 1 mole of this particular moon? Ensure your answer has the correct number of significant figures.

ANSWER: _____

189. Rewrite the following chemical equation as a balanced equation using the smallest possible whole-number stoichiometric coefficients.
- $$\text{Cu}(\text{OH})_2 + \text{HI} \rightarrow \text{H}_2\text{O} + \text{CuI}_2$$

ANSWER: _____

190. There are two steps in the extraction of copper metal from chalcocite, a copper ore. In the first step, copper (I) sulfide and oxygen gas react to form copper (I) oxide and sulfur dioxide. In the second step, copper (I) oxide and carbon react to form copper and carbon monoxide. Write the net chemical equation for the production of copper from copper (I) sulfide, oxygen, and carbon. Ensure that it is balanced.

ANSWER: _____

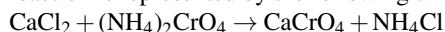
191. What is the oxidation number of sulfur in zinc sulfide?

ANSWER: _____

192. A strip of solid bismuth metal is put into a beaker of 0.010-M $\text{Pt}(\text{NO}_3)_2$ solution. Does a chemical reaction occur? Explain why it does or why it does not. If it does occur, write the net ionic equation that describes it. Be sure your equation is balanced and has physical-state symbols.

ANSWER: _____

193. A way to produce solid calcium chromate is by combining aqueous calcium chloride with aqueous ammonium chromate. The chemical reaction is represented by the following unbalanced chemical equation:



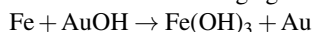
Sean wants to produce 17.1 mg of calcium chromate. He pours 8.78 mL of a 0.03352-M solution of aqueous calcium chloride into a beaker. What volume of a 0.880-M solution of aqueous ammonium chromate is needed to react with the calcium chloride solution to produce 17.1 mg of calcium chromate? If you find that 8.78 mL of the calcium chloride solution is insufficient, report the required volumes of both reactants required to produce the 17.1 mg of calcium chromate.

ANSWER: _____

194. The extraction of zinc from its sulfide ores involves a two-step roasting and reduction process. In the first step, zinc sulfide reacts with oxygen to form zinc oxide and sulfur dioxide. In the second step, zinc oxide is reduced by carbon to form zinc metal and carbon monoxide. Write the net chemical equation for the production of zinc from zinc sulfide, oxygen, and carbon. Ensure that it is balanced.

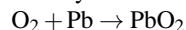
ANSWER: _____

195. Determine whether the following reaction is an oxidation-reduction (redox) reaction. If the reaction is a redox reaction, write down the formula of the reducing agent and the formula of the oxidizing agent. Provide an explanation for why you made these choices.



ANSWER: _____

196. As Pb is converted into PbO₂ in the following reaction, determine whether lead is oxidized, reduced, or neither. Clearly explain why you chose your answer. Consider only the lead found in Pb and PbO₂.



ANSWER: _____

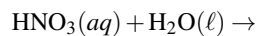
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ANSWER: _____

198. Ammonium fluoride reacts with aluminum oxide to form ammonium oxide and aluminum fluoride. If 8.044 g of ammonium oxide is produced from the reaction of 32.57 g of ammonium fluoride and 27.8 g of aluminum oxide, calculate the percent yield of ammonium oxide. Ensure that your answer has the correct number of significant figures.

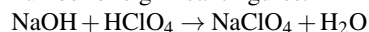
ANSWER: _____

199. Predict the products of the reaction below by completing the right-hand side of the chemical equation. Provide the full balanced equation with state symbols.



ANSWER: _____

200. Given this unbalanced chemical equation, report the mass of HClO₄ needed to produce 21.5 g of H₂O. Ensure your answer has the correct number of significant figures.



ANSWER: _____